Freezer Case Tracking: Mockup and Rules

October 17, 2026 · Frank · Rev. 2: 4 × 3 in label, 6-digit serials

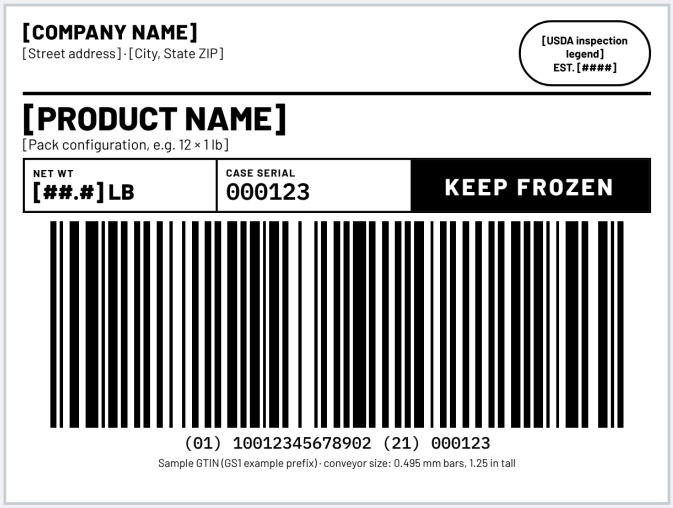
Each case gets one barcode holding its case GTIN and a unique serial number. Scan it once going into the freezer and once coming out: the scanner confirms the product and shows how many days it was frozen.

How it works

1. **Vendor preprints labels.** One roll per product, each label with its own serial number, in ranges you assign.
2. **Apply at packing.** One 4 × 3 in label per case, on the longest side.
3. **Scan in at the freezer.** The app checks the product and records the time. That time is the freezer-entry time.
4. **Scan out when pulling.** The app checks the product against the order and shows days frozen. It stops the case if the product is wrong, the case is over your limit, or it was never scanned in.
5. **Count weekly.** COUNT mode checks every case in the freezer against the app's list.

The case label

A 4 × 3 in label with one GS1-128 barcode. Nothing on it changes after printing, so the whole label can be preprinted.



One barcode, two facts

`(01) 10012345678902`

Case GTIN: which product. The same on every label in a roll. One roll per product.

`(21) 000123`

Serial: which case. 6 digits, different on every label, never reused.

Both are fixed when printed, so a vendor can **preprint** them. The scan times live in the app.

Shown at conveyor-scanner size (GS1 US): 0.495 mm bars, 1.25 in tall, 3.86 in wide with blank margins. A longer serial would not fit a 4-inch label. [Brackets] = your details. The barcode is real and scans.

Full-size, editable mockup: Freezer Scan Mockup

Scanner screens

The scanner has three modes: SCAN IN, SCAN OUT and COUNT. Every scan ends in one of the results below. Green means go. Amber means hold for review. Red means stop, and nothing is logged. Blue is the weekly count.



Weekly count

The weekly count stays, using the scanner's COUNT mode. Every case found is checked against the app's list, which also exposes missed scans.

- **COUNT never logs a case in or out.** Each case's freezer time is unchanged.
- **Pause scan-in and scan-out** in that freezer until the count is finished.
- **Labels face out** so every case can be scanned. Labels on two adjacent sides help.
- **Resolve every mismatch before the next count.** Missing means it left without a scan-out, or it's lost or mislabeled. Not expected means it was never scanned in, and a supervisor enters its in-date.
- **The app is the record of freezer inventory.** QuickBooks inventory is not used.

Rules

Labels and serials

- **One case GTIN per product**, 14 digits, separate from the retail UPC. UNFI accepts a unit UPC on cases only as a fallback.
- **Serials are 6 digits**, e.g. `000123` (999,999 cases per product). At conveyor size an 8-digit serial makes the barcode 4.07 in wide and a 10-digit one 4.29 in: too wide for a 4-inch label.
- **Barcode size (GS1 US):** 0.495 mm bars, 1.25 in tall if buyer distribution centers scan on conveyors; 0.264 mm bars, 0.5 in tall if cases are only scanned by hand. The mockup uses conveyor size.
- **4 × 3 in label:** check required text sizes (inspection legend, net weight) with your USDA inspector.
- **You assign every serial range** and log it in a range ledger before the vendor prints. A serial is never reused for the same GTIN.
- **Freezer-grade adhesive.** Barcode on the longest side, at least 1.25 in from any edge or seam (UNFI).
- **Keep a spare-label roll** with its own serial range for damaged or duplicate labels.

Scanning

- **Every case is scanned in** when it enters the freezer. No scan, no entry.
- **Every case is scanned out against an order** when it is pulled.
- **A case pulled and put back** is scanned with "Return to freezer", not a new scan-in.
- **Freezer-rated scanners**, or scan at the freezer door.

The app

- **A GTIN + serial scans in once, ever.** A repeat is blocked as a duplicate. The first record is never changed.
- **Each serial is linked to its production lot** at scan-in. GS1 US says to keep a lot reference for traceability.
- **Days frozen = scan-out time minus scan-in time.** Cases over the limit are held, not shipped.
- **New rolls are checked on arrival:** scan the first and last label. The app rejects any range that overlaps the ledger.
- **Scans queue on the scanner** when Wi-Fi drops, and keep their original time.
- **Only a supervisor can override a stop** or enter a missing in-date. Each override is logged with a name and a reason.

Vendor contract

- No duplicate serials. The vendor reprints at its own cost if any appear.
- Each shipment comes with a CSV listing the GTIN and serial range of every roll.
- Barcodes verified to ISO grade C (1.5) or better.

Decide before ordering labels

The first two decide whether labels can be fully preprinted. GS1 US guidance treats GTIN + serial as meeting the core case requirement. It appears to call for the date inside the barcode only on variable-weight meat cases.

☐ Are case weights fixed or variable? Variable weight puts weight and date in the barcode, which means printing on-site.

☐ Ask Hannaford and UNFI buyers whether they require a date or lot inside the case barcode, and whether their distribution centers scan cases on conveyors or by hand.

☐ Assign a case GTIN for each product from your GS1 company prefix.

☐ Confirm your UPCs come from a GS1 license, not a reseller.

☐ Set the freezer limit in days for each product.

☐ Choose the scanner model and the label vendor.

Sources

- North American Guidance for Standard Case Code Labeling, Release 3.0, GS1 US, Dec 2025: core case data, serial length, meat and poultry attributes.
- UNFI Packaging Guidelines, RetailerHub summary of UNFI's May 2024 requirements (third-party, not UNFI's own document): case GTIN, placement.